Speed

Chapters 8 & 9 — Supplement

A units-method companion for upper-primary problem solving

Grades 4–6 | English | ThinkAlgo editorial series

What is inside

- **Chapter 8 — Fractions of a journey (the units method).** Draw a journey as equal units, find one unit, then scale up. Theory, 2 worked examples and 20 practice problems, each with a bar model.
- **Chapter 9 — Journeys with a change of speed.** There-and-back trips, partial legs, differences in average speed, and speed changes mid-journey. Theory, 2 worked examples and 20 practice problems.

This booklet is produced in a single source with two editions.
Teacher edition. Full step-by-step solutions are shown.

Chapter 8

Fractions of a journey — the units method

Many journey problems become easy once you picture the whole trip as a row of *equal units*. If you know how far (or how long) *one* unit is, you can find the *whole* just by scaling up. That single idea powers this entire chapter.

Key idea — the units method

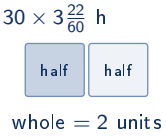
Split the journey into equal units. Choose units so the part you know is a whole number of them.

1. **Draw** the journey as equal units (a bar model).
2. **Find 1 unit** from the part you are given (for example, "7 units = 840 km, so 1 unit = 120 km").
3. **Scale up** to the whole: whole = (number of units) × (1 unit).

Two useful specials: **(a)** if you travel *half* the trip, the whole is 2× that part; **(b) average speed** $= \dfrac{\text{total distance}}{\text{total time}}$ (never just the average of the two speeds).

Example 1 — a half journey

Emma cycles for **3 h 22 min** at **30 km/h** and arrives exactly *halfway* along her route. How long is the whole route?



Solution

Write the time as hours. 3 h 22 min $= 3\frac{22}{60} = \dfrac{202}{60}$ h.

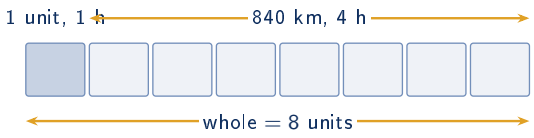
Find the half (one unit). Distance = speed × time $= 30 \times \dfrac{202}{60} = 101$ km.

Scale up to the whole. The whole route is 2 units: $2 \times 101 = 202$ km.

> **Answer:** The whole route is **202 km**.

Example 2 — units and average speed

Oliver drives $\frac{1}{8}$ **of a journey in 1 h**. The remaining **840 km** takes him **4 h**. Find the length of the whole journey and his average speed for the trip.



Solution

Count the units. The journey is 8 units; $\frac{1}{8}$ is 1 unit, the rest is 7 units.

Find 1 unit. 7 units $= 840$ km, so 1 unit $= 840 \div 7 = 120$ km.

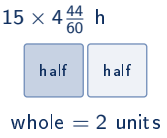
Scale up to the whole. Whole $= 8 \times 120 = 960$ km.

Average speed. Total time $= 1 + 4 = 5$ h, so average $= 960 \div 5 = 192$ km/h.

> **Answer:** Whole journey $=$ **960 km**; average speed $=$ **192 km/h**.

Practice. Full solutions follow each problem.

1 A cyclist rides for **4 h 44 min** at **15 km/h** and reaches the *halfway* point of a trail. How long is the whole trail?



Solution

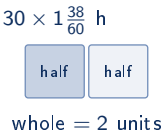
Time in hours. 4 h 44 min $= \dfrac{284}{60}$ h.

Find the half. $15 \times \dfrac{284}{60} = 71$ km.

Whole $= 2$ units. $2 \times 71 = 142$ km.

> **Answer:** **142 km**.

2 A runner covers **1 h 38 min** at **30 km/h**, which is exactly half of a race. Find the full length of the race.



Solution

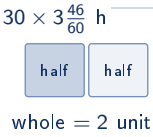
Time in hours. 1 h 38 min $= \dfrac{98}{60}$ h.

Half. $30 \times \dfrac{98}{60} = 49$ km.

Whole. $2 \times 49 = 98$ km.

> **Answer:** **98 km**.

3 A car travels for **3 h 46 min** at **30 km/h** to reach the midpoint of a highway. How long is the whole highway?



Solution

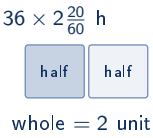
Time in hours. 3 h 46 min $= \dfrac{226}{60}$ h.

Half. $30 \times \dfrac{226}{60} = 113$ km.

Whole. $2 \times 113 = 226$ km.

Answer: **226 km**.

4 Ben drives **2 h 20 min** at **36 km/h** and that is half of his whole journey. Find the whole distance.



Solution

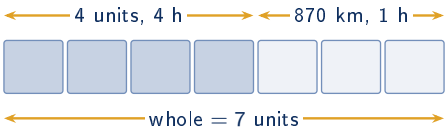
Time in hours. 2 h 20 min $= \dfrac{140}{60}$ h.

Half. $36 \times \dfrac{140}{60} = 84$ km.

Whole. $2 \times 84 = 168$ km.

Answer: **168 km**.

5 Sophia completes $\frac{4}{7}$ **of a journey in 4 h**. The remaining **870 km** takes just **1 h** on a fast road. Find the whole journey and her average speed.



Solution

Count units. 7 units in all; $\frac{4}{7}$ is 4 units, the rest is 3 units.

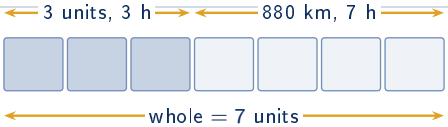
Find 1 unit. 3 units = 870, so 1 unit = 870 ÷ 3 = 290 km.

Whole. $7 \times 290 = 2030$ km.

Average speed. Total time $= 4 + 1 = 5$ h, so 2030 ÷ 5 = 406 km/h.

Answer: Whole = **2030 km**; average speed = **406 km/h**.

6 Lucas cycles $\frac{3}{7}$ **of a route in 3 h**, then the last **880 km** take **7 h**. Find the whole route and his average speed.



Solution

Count units. 7 units; $\frac{3}{7}$ is 3 units, the rest is 4 units.

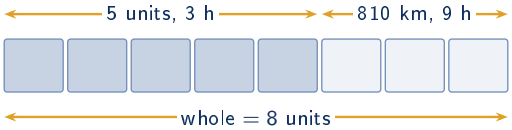
Find 1 unit. 4 units = 880, so 1 unit = 220 km.

Whole. $7 \times 220 = 1540$ km.

Average speed. Total time $= 3 + 7 = 10$ h, so $1540 \div 10 = 154$ km/h.

Answer: Whole = **1540 km**; average speed = **154 km/h**.

7 Ethan drives $\frac{5}{8}$ **of a trip in 3 h**. The remaining **810 km** take **9 h**. Find the whole trip and his average speed.



Solution

Count units. 8 units; $\frac{5}{8}$ is 5 units, the rest is 3 units.

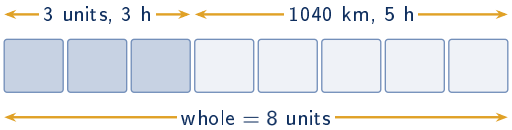
Find 1 unit. 3 units = 810, so 1 unit = 270 km.

Whole. $8 \times 270 = 2160$ km.

Average speed. Total time $= 3 + 9 = 12$ h, so $2160 \div 12 = 180$ km/h.

Answer: Whole = **2160 km**; average speed = **180 km/h**.

8 Emma covers $\frac{3}{8}$ **of a journey in 3 h**, then the last **1040 km** in **5 h**. Find the whole journey and her average speed.



Solution

Count units. 8 units; $\frac{3}{8}$ is 3 units, the rest is 5 units.

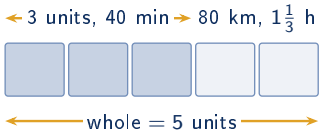
Find 1 unit. 5 units = 1040, so 1 unit = 208 km.

Whole. $8 \times 208 = 1664$ km.

Average speed. Total time $= 3 + 5 = 8$ h, so $1664 \div 8 = 208$ km/h.

Answer: Whole = **1664 km**; average speed = **208 km/h**.

9 Sophia walks $\frac{3}{5}$ **of a path in 40 min**, then the last **80 km** take $1\frac{1}{3}$ **h**. Find the whole path and her average speed.



Solution

Count units. 5 units; $\frac{3}{5}$ is 3 units, the rest is 2 units.

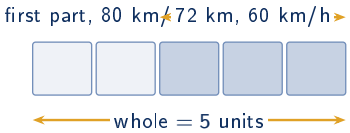
Find 1 unit. 2 units = 80, so 1 unit = 40 km.

Whole. $5 \times 40 = 200$ km.

Average speed. Total time $= \frac{40}{60} + 1\frac{1}{3} = \frac{2}{3} + \frac{4}{3} = 2$ h, so $200 \div 2 = 100$ km/h.

> **Answer:** Whole = **200 km**; average speed = **100 km/h**.

10 Ben drives from town A to town B. The *last* $\frac{3}{5}$ of the whole trip is **72 km**, driven at **60 km/h**. He drives the first part at **80 km/h**. Find the length of the first part and the total time.



Solution

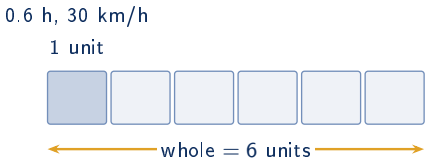
Find 1 unit. The last $\frac{3}{5}$ is 3 units = 72 km, so 1 unit = 24 km.

Whole and first part. Whole $= 5 \times 24 = 120$ km; first part = 2 units = 48 km.

Total time. $48 \div 80 + 72 \div 60 = 0.6 + 1.2 = 1.8$ h.

> **Answer:** First part = **48 km**; total time = **1.8 h**.

11 A train covers $\frac{1}{6}$ **of a line in 0.6 h** at **30 km/h**. How long is the whole line?



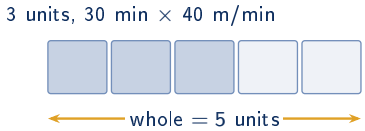
Solution

Find 1 unit. 1 unit $= 30 \times 0.6 = 18$ km.

Whole. $6 \times 18 = 108$ km.

> **Answer:** **108 km**.

12 Emma walks for **30 min** at **40 m/min**, which is $\frac{3}{5}$ of her way to school. How far is school (in metres)?



Solution

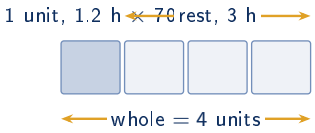
Distance walked. $40 \times 30 = 1200$ m, and that is 3 units.

Find 1 unit. $1200 \div 3 = 400$ m.

Whole. $5 \times 400 = 2000$ m.

Answer: **2000 m** (2 km).

13 Bus A drives for **1.2 h at 70 km/h**, covering $\frac{1}{4}$ of a route. Bus B then drives the rest in **3 h**. Find the whole route and the average speed for the trip.



Solution

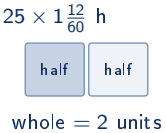
Find 1 unit. 1 unit $= 70 \times 1.2 = 84$ km.

Whole. $4 \times 84 = 336$ km.

Average speed. Total time $= 1.2 + 3 = 4.2$ h, so $336 \div 4.2 = 80$ km/h.

Answer: Whole = **336 km**; average speed = **80 km/h**.

14 A scooter rides **1 h 12 min** at **25 km/h** to reach the halfway point of a lane. How long is the whole lane?



Solution

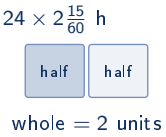
Time in hours. 1 h 12 min $= \dfrac{72}{60} = 1.2$ h.

Half. $25 \times 1.2 = 30$ km.

Whole. $2 \times 30 = 60$ km.

Answer: **60 km**.

15 A hiker walks **2 h 15 min** at **24 km/h** (on wheels!) to reach the midpoint of a trail. Find the whole length.



Solution

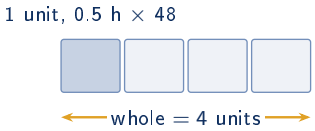
Time in hours. 2 h 15 min = 2.25 h.

Half. $24 \times 2.25 = 54$ km.

Whole. $2 \times 54 = 108$ km.

Answer: 108 km.

16 A car covers $\frac{1}{4}$ **of a journey in 0.5 h** at **48 km/h**. How long is the whole journey?



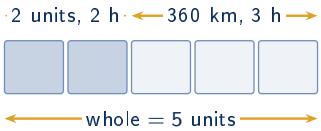
Solution

Find 1 unit. 1 unit $= 48 \times 0.5 = 24$ km.

Whole. $4 \times 24 = 96$ km.

Answer: 96 km.

17 Lucas drives $\frac{2}{5}$ **of a journey in 2 h**, then the last **360 km** in **3 h**. Find the whole journey and his average speed.



Solution

Count units. 5 units; $\frac{2}{5}$ is 2 units, the rest is 3 units.

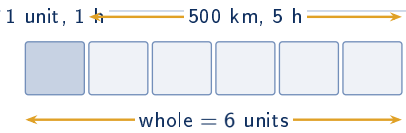
Find 1 unit. 3 units = 360, so 1 unit = 120 km.

Whole. $5 \times 120 = 600$ km.

Average speed. Total time $= 2 + 3 = 5$ h, so $600 \div 5 = 120$ km/h.

Answer: Whole = **600 km**; average speed = **120 km/h**.

18 A coach covers $\frac{1}{6}$ **of a tour in 1 h**, then the remaining **500 km** in **5 h**. Find the whole tour and the average speed.



Solution

Count units. 6 units; the rest is 5 units.

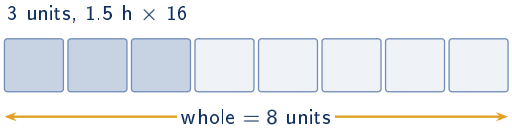
Find 1 unit. 5 units = 500, so 1 unit = 100 km.

Whole. $6 \times 100 = 600$ km.

Average speed. Total time $= 1 + 5 = 6$ h, so $600 \div 6 = 100$ km/h.

> **Answer:** Whole = **600 km**; average speed = **100 km/h**.

19 A train covers $\frac{3}{8}$ **of a line in 1.5 h** at **16 km/h**. How long is the whole line?



Solution

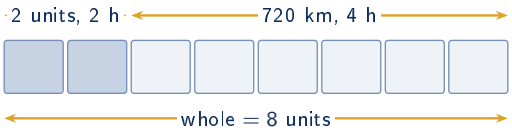
Distance covered. $16 \times 1.5 = 24$ km $= 3$ units.

Find 1 unit. $24 \div 3 = 8$ km.

Whole. $8 \times 8 = 64$ km.

> **Answer:** **64 km**.

20 Oliver drives $\frac{1}{4}$ **of a journey in 2 h**, then the remaining **720 km** in **4 h**. Find the whole journey and his average speed.



Solution

Count units. Take the journey as 8 units; $\frac{1}{4} = 2$ units, the rest is 6 units.

Find 1 unit. 6 units = 720, so 1 unit = 120 km.

Whole. $8 \times 120 = 960$ km.

Average speed. Total time $= 2 + 4 = 6$ h, so $960 \div 6 = 160$ km/h.

> **Answer:** Whole = **960 km**; average speed = **160 km/h**.

Chapter 9

Journeys with a change of speed

Real trips rarely keep one speed. A driver slows for traffic, speeds up on a clear road, or returns home more slowly than they set out. The trick is always the same: *break the journey into legs*, handle each leg with the speed triangle, then combine. Keep distance, speed and time in matching units, and remember that a *round trip* covers the same distance each way.

Key idea — one leg at a time

Use the three forms of the speed relationship, whichever fits the leg you are on:

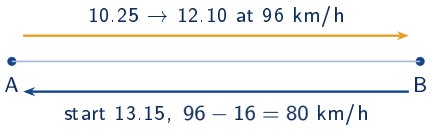
$$\text{distance} = \text{speed} \times \text{time}, \quad \text{time} = \frac{\text{distance}}{\text{speed}}, \quad \text{speed} = \frac{\text{distance}}{\text{time}}.$$

1. **Split** the trip into legs (out / back, slow / fast, first part / rest).
2. **Fill the gaps** leg by leg — a small table keeps distance, speed and time tidy.
3. **Combine.** For a there-and-back trip the two distances are equal. For an average, use $\text{average} = \dfrac{\text{total distance}}{\text{total time}}$.

Write clock times with a **dot**: 13.00, not "13 00". A "20% slower" speed means multiply by 0.8; "20% faster" means multiply by 1.2.

Example 1 — there and back

Ben drives from home (A) to a lake (B), leaving at **10.25** and arriving at **12.10** at **96 km/h**. After a rest he starts back at **13.15**, but traffic makes him **16 km/h** slower. At what time does he reach home?



Solution

Length of the trip (the "out" leg). From 10.25 to 12.10 is 1 h 45 min = 1.75 h, so $AB = 96 \times 1.75 = 168$ km.

Return speed. $96 - 16 = 80$ km/h.

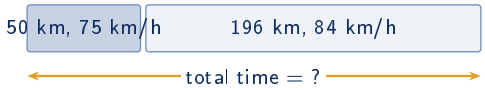
Return time. $168 \div 80 = 2.1$ h = 2 h 6 min.

Arrival time. 13.15 + 2 h 6 min = 15.21.

> **Answer:** He reaches home at **15.21**.

Example 2 — two legs, two speeds

Lucas drives the first **50 km at 75 km/h**, then another **196 km at 84 km/h**. How long does the whole journey take?



Solution

First leg. $50 \div 75 = \frac{2}{3}$ h $= 40$ min.

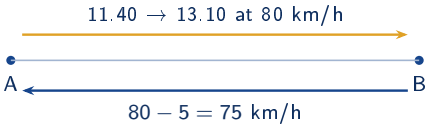
Second leg. $196 \div 84 = \frac{7}{3}$ h $= 2$ h 20 min.

Add the legs. 40 min $+ 2$ h 20 min $= 3$ h.

> **Answer:** The whole journey takes **3 h**.

Practice. Full solutions follow each problem.

1 Lucas leaves at **11.40** and arrives at **13.10** driving at **80 km/h**. He returns along the same road **5 km/h** slower. How long does the return journey take?



Solution

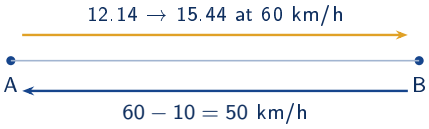
Distance. 11.40 to 13.10 is 1.5 h, so $AB = 80 \times 1.5 = 120$ km.

Return speed. $80 - 5 = 75$ km/h.

Return time. $120 \div 75 = 1.6$ h $= 1$ h 36 min.

> **Answer:** The return takes **1.6 h** (1 h 36 min).

2 Emma leaves at **12.14** and arrives at **15.44** at **60 km/h**. Her return is **10 km/h** slower. How long does the return take?



Solution

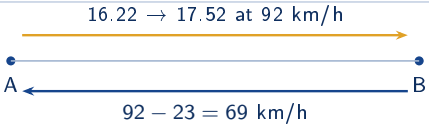
Distance. 12.14 to 15.44 is 3.5 h, so $AB = 60 \times 3.5 = 210$ km.

Return speed. $60 - 10 = 50$ km/h.

Return time. $210 \div 50 = 4.2$ h $= 4$ h 12 min.

> **Answer:** The return takes **4.2 h** (4 h 12 min).

3 Oliver leaves at **16.22** and arrives at **17.52** at **92 km/h**. He comes back **23 km/h** slower. How long is the return journey?



Solution

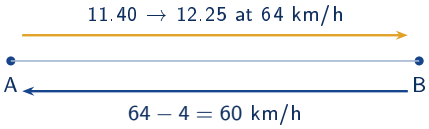
Distance. 16.22 to 17.52 is 1.5 h, so $AB = 92 \times 1.5 = 138$ km.

Return speed. $92 - 23 = 69$ km/h.

Return time. $138 \div 69 = 2$ h.

Answer: The return takes **2 h**.

4 Sophia leaves at **11.40** and arrives at **12.25** at **64 km/h**. She returns **4 km/h** slower. How long does the return take?



Solution

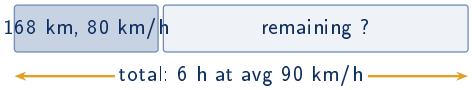
Distance. 11.40 to 12.25 is 45 min = 0.75 h, so $AB = 64 \times 0.75 = 48$ km.

Return speed. $64 - 4 = 60$ km/h.

Return time. $48 \div 60 = 0.8$ h $= 48$ min.

Answer: The return takes **0.8 h** (48 min).

5 Emma plans a **6 h** drive at an average of **90 km/h**. She drives the first **168 km at 80 km/h**. At what speed must she cover the rest to keep her planned average? (1 d.p.)



Solution

Total distance. $90 \times 6 = 540$ km.

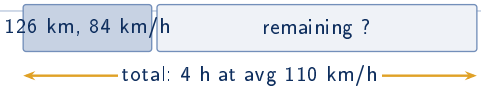
First leg's time. $168 \div 80 = 2.1$ h.

What is left. Distance $540 - 168 = 372$ km in time $6 - 2.1 = 3.9$ h.

Required speed. $372 \div 3.9 = 95.38\ldots \approx 95.4$ km/h.

Answer: About **95.4 km/h**.

6 Ben drives for **4 h** at an average of **110 km/h**. The first **126 km** are at **84 km/h**. Find the speed needed for the rest. (1 d.p.)



Solution

Total distance. $110 \times 4 = 440$ km.

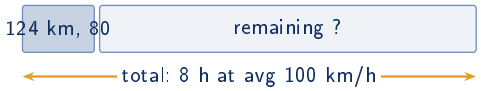
First leg's time. $126 \div 84 = 1.5$ h.

What is left. $440 - 126 = 314$ km in $4 - 1.5 = 2.5$ h.

Required speed. $314 \div 2.5 = 125.6$ km/h.

Answer: **125.6 km/h**.

7 Ethan drives for **8 h** at an average of **100 km/h**. The first **124 km** are at **80 km/h**. Find the speed needed for the rest. (1 d.p.)



Solution

Total distance. $100 \times 8 = 800$ km.

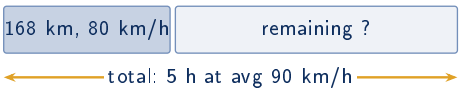
First leg's time. $124 \div 80 = 1.55$ h.

What is left. $800 - 124 = 676$ km in $8 - 1.55 = 6.45$ h.

Required speed. $676 \div 6.45 = 104.80\ldots \approx 104.8$ km/h.

Answer: About **104.8 km/h**.

8 Sophia drives for **5 h** at an average of **90 km/h**. The first **168 km** are at **80 km/h**. Find the speed needed for the rest. (1 d.p.)



Solution

Total distance. $90 \times 5 = 450$ km.

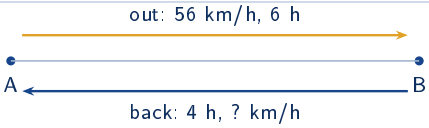
First leg's time. $168 \div 80 = 2.1$ h.

What is left. $450 - 168 = 282$ km in $5 - 2.1 = 2.9$ h.

Required speed. $282 \div 2.9 = 97.24\ldots \approx 97.2$ km/h.

Answer: About **97.2 km/h**.

9 Oliver drives to a city in **6 h** at **56 km/h**, then returns along the same road in **4 h**. By how much do his two average speeds differ?



Solution

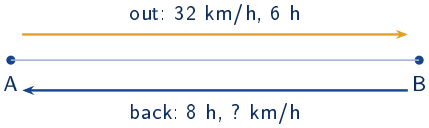
Distance. $56 \times 6 = 336$ km (the same each way).

Return speed. $336 \div 4 = 84$ km/h.

Difference. $84 - 56 = 28$ km/h.

Answer: They differ by **28 km/h**.

10 Ben drives out in **6 h** at **32 km/h** and comes back in **8 h**. By how much do his average speeds differ?



Solution

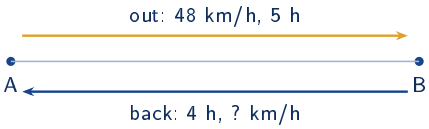
Distance. $32 \times 6 = 192$ km.

Return speed. $192 \div 8 = 24$ km/h.

Difference. $32 - 24 = 8$ km/h.

Answer: They differ by **8 km/h**.

11 Emma drives out in **5 h** at **48 km/h** and returns in **4 h**. Find the difference in her average speeds.



Solution

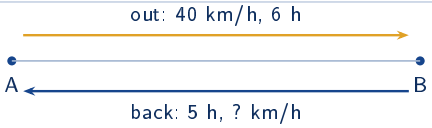
Distance. $48 \times 5 = 240$ km.

Return speed. $240 \div 4 = 60$ km/h.

Difference. $60 - 48 = 12$ km/h.

Answer: They differ by **12 km/h**.

12 Lucas drives out in **6 h** at **40 km/h** and returns in **5 h**. Find the difference in his average speeds.



Solution

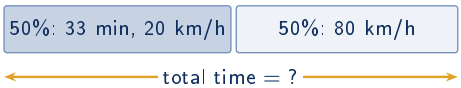
Distance. $40 \times 6 = 240$ km.

Return speed. $240 \div 5 = 48$ km/h.

Difference. $48 - 40 = 8$ km/h.

Answer: They differ by **8 km/h**.

13 Sophia covers the **first 50%** of a trip in **33 min at 20 km/h**, then drives the rest at **80 km/h**. Find the total time. (1 d.p., in minutes)



Solution

First half's distance. $20 \times \frac{33}{60} = 11$ km. (So the whole trip is 22 km.)

Second half's time. Same 11 km at 80: $11 \div 80 = 0.1375$ h $= 8.25$ min.

Total. $33 + 8.25 = 41.25 \approx 41.3$ min.

Answer: About **41.3 min**.

14 Ethan covers the **first 50%** of a trip in **9 min at 40 km/h**, then the rest at **160 km/h**. Find the total time. (1 d.p., in minutes)



Solution

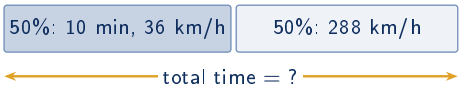
First half's distance. $40 \times \frac{9}{60} = 6$ km.

Second half's time. $6 \div 160 = 0.0375$ h $= 2.25$ min.

Total. $9 + 2.25 = 11.25 \approx 11.3$ min.

Answer: About **11.3 min**.

15 Emma covers the **first 50%** of a trip in **10 min at 36 km/h**, then the rest at **288 km/h** (a fast train!). Find the total time. (1 d.p., in minutes)



Solution

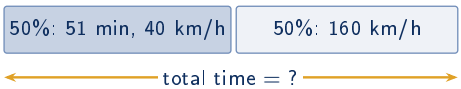
First half's distance. $36 \times \frac{10}{60} = 6$ km.

Second half's time. $6 \div 288 = 0.0208\ldots$ h $= 1.25$ min.

Total. $10 + 1.25 = 11.25 \approx 11.3$ min.

Answer: About **11.3 min**.

16 Oliver covers the **first 50%** of a trip in **51 min at 40 km/h**, then the rest at **160 km/h**. Find the total time. (1 d.p., in minutes)



Solution

First half's distance. $40 \times \frac{51}{60} = 34$ km.

Second half's time. $34 \div 160 = 0.2125$ h $= 12.75$ min.

Total. $51 + 12.75 = 63.75 \approx 63.8$ min.

Answer: About **63.8 min**.

17 Ethan drives the first $\frac{5}{9}$ of a journey at **80 km/h**. For the last $\frac{4}{9}$ he slows by **20%**, which takes him **2.5 h**. Find the total time for the journey.



Solution

Speed for the last part. 20% slower $= 80 \times 0.8 = 64$ km/h.

Length of the last part. $64 \times 2.5 = 160$ km $= \frac{4}{9}$ of the whole, so $\frac{1}{9} = 40$ km and the whole $= 360$ km.

First part. $\frac{5}{9} = 200$ km at 80 km/h $= 200 \div 80 = 2.5$ h.

Total time. $2.5 + 2.5 = 5$ h.

Answer: The journey takes **5 h**.

18 Emma drives the first $\frac{5}{9}$ of a journey at **80 km/h**. For the last $\frac{4}{9}$ she speeds up by **20%**, which takes her **2.5 h**. Find the total time for the journey.



Solution

Speed for the last part. 20% faster $= 80 \times 1.2 = 96$ km/h.

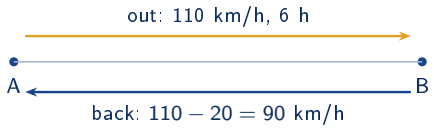
Length of the last part. $96 \times 2.5 = 240$ km $= \frac{4}{9}$ of the whole, so $\frac{1}{9} = 60$ km and the whole $= 540$ km.

First part. $\frac{5}{9} = 300$ km at 80 km/h $= 300 \div 80 = 3.75$ h.

Total time. $3.75 + 2.5 = 6.25$ h $= 6$ h 15 min.

> **Answer:** The journey takes **6.25 h** (6 h 15 min).

19 A coach drives for **6 h at 110 km/h** to a resort, then returns along the same road **20 km/h** slower. How long does the return journey take?



Solution

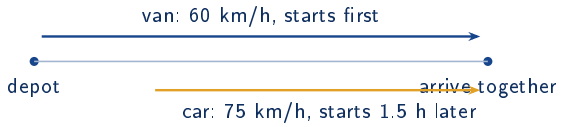
Distance. $110 \times 6 = 660$ km.

Return speed. $110 - 20 = 90$ km/h.

Return time. $660 \div 90 = \frac{22}{3}$ h $= 7$ h 20 min.

> **Answer:** $\frac{22}{3}$ **h** $=$ **7 h 20 min**.

20 A van leaves a depot at **60 km/h**. A car sets off along the same road **1.5 h later** at **75 km/h**, and the two arrive at the destination at the same moment. For how long does the *car* travel?



Solution

Let the car travel t hours. Then the van travels $t + 1.5$ hours.

Same distance. $60(t + 1.5) = 75t$.

Solve. $60t + 90 = 75t \Rightarrow 90 = 15t \Rightarrow t = 6$ h. (Check: both cover $75 \times 6 = 450$ km; the van takes 7.5 h.)

> **Answer:** The car travels for **6 h**.

End of supplement

ThinkAlgo • Speed • Chapters 8 & 9

Teacher edition